

Camera preparation

1. Mount camera on the iPad.



We do not have a bracket dedicated to our iPad. The simplest solution is to attach it using adhesive tape. It would be great to buy a proper bracket or 3D print it.

2. Calibrate the camera.



The camera needs to be calibrated after every montage using [Structure Sensor Calibrator app](#). Ideally, it should be performed outdoors. It can be calibrated indoors using printed out [indoor calibration target](#), but it still needs some daylight.

3. If possible, mount the iPad on a tall stable stand.

Makes it easier to perform a scan, especially for short personnel.

Patient preparation

1. Adhere ECGI electrodes to the patient's body.
2. Lay the patient supine. Make sure the facial hair does not cover the torso.
3. Place the patient in good lighting conditions.
4. Lower the bed as possible.

Scanning

1. Run [Structure Capture app](#).



The app requires paid subscription (20 usd / month or 200 usd / year). Recently I used 1 week free trial. With the camera comes a 1 year free subscription (the code that I got did not work last time but I hope it will work next time).

There are some third-party alternatives. Some of them are paid, some free, but inconvenient for our use case, so I would prefer to use the dedicated app.

2. Adjust camera settings.
 - a. Scanning area ~ 1m x 1m x 1m
 - b. Scan offset ~ 0.8 m
 - c. Scan resolution < 1 mm
 - d. Scan preset: face/body



The camera sometimes disconnects from the iPad. Then, it has to be reconnected and some of the settings reset.

3. Place the camera ~1m above the patient's torso and start scanning.



Hold the camera around 5s seconds directly above the torso. Then, if the sides of the torso are not captured, move the camera a bit to the sides, holding a few seconds in each position. Stop scanning before taking down the iPad.

4. Render the RGB view (may turn out unnecessary in the further research).
5. Export the result, so that it can be used on the computer running the reconstruction script.